Reduction #9: The Gabor-kernel Tempotron
Band-pass Radiality and the Two-Quadrature Capacity Bound

Tempotron learning-surface program

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Abstract

We treat the Gabor (band-pass) post-synaptic-potential (PSP) kernel $K(s) = e^{-s^2/2\sigma^2}\cos(\omega_0 s)$ as a stand-alone reduction: the program's *radiality axis*. Its quality factor $Q = \omega_0\sigma$ interpolates continuously from the low-pass Gaussian ($Q \to 0$) to a single carrier frequency ($Q \to \infty$); the effective dimension is pinned exactly by the spectral moments,

$$K_{\text{eff}} \;=\; T\sqrt{\lambda_2/\lambda_0} \;=\; T\sqrt{\tfrac{1}{2\sigma^2} + \frac{\omega_0^2}{1+e^{-Q^2}}} \;\xrightarrow{Q\to\infty}\; \omega_0 T\sqrt{1+\tfrac{1}{2Q^2}}.$$

Our main contribution is theoretical. In the narrowband limit the membrane potential collapses to a single quadrature pair, $V(t) = A\cos\omega_0 t + B\sin\omega_0 t$ with $A = \mathbf{w}\cdot\mathbf{a}$, $B = \mathbf{w}\cdot\mathbf{b}$, so that $\max_t V = \sqrt{A^2+B^2}$ and the tempotron decision becomes the *two-quadrature radial classifier* $\sqrt{(\mathbf{w}\cdot\mathbf{a})^2 + (\mathbf{w}\cdot\mathbf{b})^2} > \theta$. Solving the replica-symmetric (RS) Gardner distance-to-feasible-set problem for this geometry gives a closed form,

$$\frac{1}{\alpha_c(\theta)} \;=\; \tfrac{1}{2}\big(\theta^2 - \sqrt{2\pi}\,\theta + 2\big),$$

which at the balanced (median) threshold $\theta = \sqrt{2\ln 2}$ yields $\boxed{\alpha_c^{\text{RS}} = 4.598}$ (4.660 threshold-optimized). This is, to our knowledge, a *new* value: the two-quadrature norm-threshold capacity is unpublished, and it is decisively distinct from the literature one-quadrature squared-perceptron value $\alpha_c = 4$ (Gratsea–Kasper–Lewenstein; Cruciani). We verified it four independent ways (a Gardner min-cost calculation, a SymPy closed form, an 8 M-sample Monte-Carlo 4.601, and an independent codex re-derivation), all anchored to the exact linear value 2. Because the fire region is the *exterior of a disk*—non-convex—RSB is permitted, so 4.598 is an RS/existence value and the findable capacity is lower. On the experimental side: the narrowband Gabor crosses at $\hat\alpha_c \approx 3.55$ (the radial regime, findable well below the RS 4.6, consistent with the single-sinusoid reduction #12 at 3.4–3.8), and the decisive *matched-K_{eff}* band-pass>low-pass isolation—bugged in the first campaign by a one-line argv error that ignored the carrier frequency and railed every cell to $K_{\text{eff}} \approx 499$—is now **CONFIRMED** with the fix: at matched K_{eff} the radiality knob κ_S (3→2, low-pass→band-pass) raises $\hat\alpha_c$ by $\approx +0.1$ (3.15→3.25 at K_{eff}=32, band-pass higher at K_{eff}=64), directly corroborating the two-axis finding (reduction #7), with the advantage growing as κ_S→1. Status: **CONFIRMED**—the radial theory is solid and quadruply verified, and the empirical matched-K_{eff} control now isolates the band-pass advantage (sign unambiguous; $\frac{1}{2}$-crossing precision ±0.1, log-derived).

Contents

Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in `lit/tempotron-reductions/derivations/`.

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1 Introduction

The tempotron [8] fires when its peak membrane potential exceeds threshold,

$$V_{\max}(\mathbf{w}) \;=\; \max_{t\in[0,T]} \sum_{i=1}^{N} w_i \sum_f K(t - t_i^f), \tag{1}$$

where $K(s)$ is the PSP kernel and $\{t_i^f\}$ the presynaptic spikes. The companion reductions in this series isolate one structural property of K at a time: smoothness (#8 Gaussian, #11 rectangular), causality (#10 alpha), band-limitedness (#6 sinusoidal). The present paper isolates *radiality*: what happens to capacity when the kernel carries a frequency, so the readout sees not a single linear projection but the *magnitude* of a two-dimensional projection.

Why the Gabor kernel. The Gabor elementary signal [5]

$$K(s) \;=\; e^{-s^2/2\sigma^2}\cos(\omega_0 s), \qquad Q \;:=\; \omega_0\sigma, \tag{2}$$

is the minimal kernel whose quality factor Q dials *continuously* between two qualitatively different readouts. At $Q \to 0$ the cosine flattens and $K \to e^{-s^2/2\sigma^2}$, the pure low-pass Gaussian of reduction #8: a smooth broadband process read out by a best-of-many-looks linear perceptron. At $Q \to \infty$ the envelope is flat over any finite spike window and $K \to \cos\omega_0 s$, a single carrier: the readout becomes the amplitude of one in-phase/quadrature pair, a categorically non-linear decision. The Gabor kernel is therefore the one that *breaks* a naive "α_c is a function of K_{eff} alone" universality and forces a two-axis picture (effective dimension × radiality, the subject of the cross-kernel reduction #7).

Contribution. We (i) write the exact narrowband reduction of the tempotron decision to a two-quadrature radial classifier (§2); (ii) *solve* the RS capacity of that classifier in closed form and obtain a new value $\alpha_c^{\text{RS}} = 4.598$, distinguishing it sharply from the literature one-quadrature $\alpha_c = 4$ and verifying it four ways (§3); (iii) report the findable lab capacity ($\hat{\alpha}_c \approx 3.55$, well below the

RS value, as the non-convexity predicts) and the now-corrected matched-$K_{\rm eff}$ control that confirms band-pass>low-pass (§4); and (iv) lay out the honest status—what is RS-level, where RSB and finite-window anisotropy enter, and what remains open (§5).

2 Model and Setup

Model. N afferents emit spike trains over $[0, T]$; the membrane potential is (1) with the Gabor kernel (2). Classification is $V_{\max} > \theta \Rightarrow$ fire, on balanced random ± 1 labels; the capacity $\alpha_c = P/N$ is the half-solving load. The linear anchor is Cover's $\alpha_c = 2$ [3, 6, 10].

Assumptions ledger.

(A1) Gaussian field. For $N \gg 1$ afferents the CLT makes $V(t)$ a stationary zero-mean Gaussian process with covariance $\mathcal{C}(\tau) = (K \star K)(\tau)$ and spectrum $S(\omega) = |\hat{K}(\omega)|^2$ (Wiener–Khinchin). This is the same $N \gg 1$ premise as reductions #3,#4,#6; it can break under sparse input / finite-N starvation.

(A2) Narrowband reduction. As $Q \to \infty$ the decision statistic $\max_t V$ reduces *exactly* to the radial form $\sqrt{(\mathbf{w}\cdot\mathbf{a})^2 + (\mathbf{w}\cdot\mathbf{b})^2} > \theta$ (§3.1); at finite Q extra modes and the envelope make the two quadratures correlated and unequal-power (§3.5).

(A3) Isotropic quadratures. In that limit the two quadrature fields $h_1 = \mathbf{w}\cdot\mathbf{a}/\sqrt{N}$, $h_2 = \mathbf{w}\cdot\mathbf{b}/\sqrt{N}$ are i.i.d. $\mathcal{N}(0, 1)$ for fixed $\mathbf{w}$ on the sphere (uniform phase, $\omega_0 T \gg 1$). Finite T / non-uniform phase introduces an anisotropy $\rho > 0$ (§3.5, the load-bearing finite-size correction).

(A4) RS Gardner min-cost. We use the RS, $q \to 1$ (zero-error) distance-to-feasible-set capacity (§3.2); it is *exact at the linear anchor* (reproduces 2). The exterior-of-disk region is non-convex, so RSB is permitted and would *lower* α_c.

(A5) Balanced threshold. θ is the median of $\max_t V$ (per-pattern fire probability $\frac{1}{2}$), the operating point the lab uses and the convention under which the $\ln \ln K$ cluster argument of reduction #3 holds. A *margin-κ* convention (the literature $\alpha_c = 4$) is a *different* problem (§3.3).

Every *absolute* α_c below is RS-level; the robust outputs are the *value* of the radial limit (and its provenance), the *sign* of the band-pass advantage, and the *shape* of $\alpha_c(Q)$.

Exact effective dimension. The real Gabor kernel (2) has a power spectrum $|\hat{K}(\omega)|^2$ made of two Gaussian lobes at $\pm\omega_0$ of width $1/\sigma$ whose tails overlap. Carrying the $\pm\omega_0$ interference cross-term through the spectral moments gives the *exact* ratio (writing $r = e^{-Q^2}$ for the residual lobe overlap)

$$\frac{\lambda_2}{\lambda_0} = \frac{1}{2\sigma^2} + \frac{\omega_0^2}{1 + e^{-Q^2}} \quad \Longrightarrow \quad \boxed{K_{\rm eff} = T\sqrt{\frac{1}{2\sigma^2} + \frac{\omega_0^2}{1 + e^{-Q^2}}} = \omega_0 T \sqrt{\frac{1}{2Q^2} + \frac{1}{1 + r}}.} \tag{3}$$

Dropping the $\pm\omega_0$ interference cross-term—i.e. treating the spectrum as an incoherent two-Gaussian power mixture, equivalently a single analytic-signal lobe—replaces $1/(1 + e^{-Q^2})$ by 1 and yields the familiar large-Q short form

$$\frac{\lambda_2}{\lambda_0} \approx \omega_0^2 + \frac{1}{2\sigma^2} \quad \Longrightarrow \quad K_{\text{eff}} \approx \omega_0 T\sqrt{1 + \tfrac{1}{2Q^2}}, \tag{4}$$

which *over*-counts λ_2/λ_0 by exactly $\omega_0^2/(e^{Q^2}+1)$ (+1.6% at $Q = 2$, +21.8% at $Q = 1$, vanishing as $Q \to \infty$). A cross-kernel comparison must therefore match cells on the *exact* axis (3), or it inherits a Q-dependent bias that would mimic a radiality effect; the short form (4) must not be substituted at low cycle counts ($Q \lesssim 1.5$). If the simulations build K_{eff} from the *numerical* spectrum of the discretized kernel, the measured K_{eff} axis is unaffected and only this closed-form prose changes. The Gaussian ($Q \to 0$) limit recovers $K_{\text{eff}} = T/(\sigma\sqrt{2})$ of reduction #8.

3 Theory: The Two-Quadrature Radial Classifier

This section is the main contribution.

3.1 Exact narrowband reduction

As $Q \to \infty$ the envelope $e^{-s^2/2\sigma^2}$ is flat across any finite spike window, and $K(s) \to \cos\omega_0 s = \cos\omega_0 t\, \cos\omega_0 t_s + \sin\omega_0 t\, \sin\omega_0 t_s$. Substituting into (1) and collecting terms,

$$V(t) \longrightarrow A\cos\omega_0 t + B\sin\omega_0 t, \qquad A = \mathbf{w}\cdot\mathbf{a},\ \ B = \mathbf{w}\cdot\mathbf{b}, \tag{5}$$

with the cos/sin quadrature features

$$a_i = \sum_f \cos\omega_0 t_i^f, \qquad b_i = \sum_f \sin\omega_0 t_i^f. \tag{6}$$

A single sinusoid attains its maximum over t at its amplitude, so

$$\max_t V(t) = \sqrt{A^2 + B^2} = \sqrt{(\mathbf{w}\cdot\mathbf{a})^2 + (\mathbf{w}\cdot\mathbf{b})^2}. \tag{7}$$

This identity is exact once the observation window spans at least one full carrier period ($\omega_0 T \gtrsim 2\pi$); for sub-period windows the maximum is attained at an endpoint and (7) holds only approximately, with boundary corrections (mirroring reduction #12). Hence the firing rule is

$$\text{fire} \iff (\mathbf{w}\cdot\mathbf{a})^2 + (\mathbf{w}\cdot\mathbf{b})^2 > \theta^2 \iff \mathbf{w}^\top M\,\mathbf{w} > \theta^2, \quad M = \mathbf{a}\mathbf{a}^\top + \mathbf{b}\mathbf{b}^\top \text{ (rank 2)}. \tag{8}$$

The decision surface in weight space is a quadric—the *exterior* of a rugby-ball ellipsoid—categorically non-linear. Because M is rank-2 rather than a free $N \times N$ form, the classifier keeps only N free parameters: its capacity is $O(1)$, not the extensive-in-N capacity of a full quadratic gate (Baldi–Vershynin [1]). Geometrically the decision lives in the $(A, B) = (\mathbf{w}\cdot\mathbf{a}, \mathbf{w}\cdot\mathbf{b})$ plane: fire is the exterior of a disk of radius θ, silent its interior (Fig. 1a).

3.2 RS Gardner distance-to-feasible-set capacity

We compute the zero-error RS capacity in the $q \to 1$ limit through the Gardner "min-cost" representation: the inverse capacity equals the expected squared distance from a frozen Gaussian field to the label-allowed region,

$$\frac{1}{\alpha_c} = \mathbb{E}_{\sigma,\mathbf{z}}\Big[\min_{\mathbf{h}\in A_\sigma} \|\mathbf{h}-\mathbf{z}\|^2\Big], \qquad \mathbf{z} \sim \mathcal{N}(0, I_d), \tag{9}$$

with d the field dimension and A_σ the region allowed for label σ. This representation reduces the replica calculation to a single geometric expectation and is *exact* at the linear anchor.

Anchor ($d = 1$, linear, $\kappa = 0$). Here $A_+ = \{h > 0\}$, $A_- = \{h < 0\}$, and the nearest allowed point of a wrong-sign field is the origin, so the cost is h^2 on the wrong half-line. Averaging the balanced labels, $1/\alpha_c = \frac{1}{2}\mathbb{E}[z^2] = \frac{1}{2}$, giving $\alpha_c = 2$ [6]. (SymPy and MC both return 2.000.)

Radial classifier ($d = 2$, ideal narrowband). Now $\mathbf{z} = (z_1, z_2) \sim \mathcal{N}(0, I_2)$ and $r = \|\mathbf{z}\|$ is Rayleigh, $p(r) = r\, e^{-r^2/2}$. The fire region $A_+ = \{\|\mathbf{h}\| \geq \theta\}$ is the exterior of a disk, the silent region $A_- = \{\|\mathbf{h}\| \leq \theta\}$ its interior. The nearest-point projection onto either region is *radial*, so the cost charged on the wrong side of a label is $(r - \theta)^2$; averaging the two balanced labels tiles $[0, \infty)$ and gives

$$\boxed{\frac{1}{\alpha_c(\theta)} \;=\; \frac{1}{2}\int_0^\infty (r-\theta)^2\, r\, e^{-r^2/2}\, dr \;=\; \frac{1}{2}\big(\theta^2 - \sqrt{2\pi}\,\theta + 2\big).} \tag{10}$$

The closed form uses the Rayleigh moments $\int_0^\infty r\, e^{-r^2/2}\, dr = 1$, $\int_0^\infty r^2 e^{-r^2/2}\, dr = \sqrt{\pi/2}$, $\int_0^\infty r^3 e^{-r^2/2}\, dr = 2$.

The balanced and optimal thresholds. The Rayleigh survival is $\mathbb{P}(R > \theta) = e^{-\theta^2/2}$, so the balanced (median) threshold with per-pattern fire probability $\frac{1}{2}$ is

$$\theta_{\text{bal}} = \sqrt{2\ln 2} \approx 1.1774 \implies \frac{1}{\alpha_c} = 1 + \ln 2 - \sqrt{\pi \ln 2} = 0.21748 \implies \boxed{\alpha_c^{\text{RS}} = 4.598.} \tag{11}$$

Minimizing (10) over θ gives $\mathrm{d}(1/\alpha_c)/\mathrm{d}\theta = \theta - \sqrt{\pi/2} = 0$, i.e. $\theta^* = \sqrt{\pi/2} \approx 1.2533$ and $1/\alpha_c = 1 - \pi/4$, so

$$\alpha_c^{\text{RS}}(\theta^*) = \frac{1}{1 - \pi/4} = 4.660: \tag{12}$$

the median operating point is within 1.5% of capacity-optimal. Figure 1b plots $\alpha_c(\theta)$ with both thresholds marked.

3.3 This is a new value, distinct from the literature $\alpha_c = 4$

The rigorous-replica quadratic-perceptron capacity $\alpha_c = 4$ [4, 7] is for the *one-quadrature* squared perceptron $\zeta = \Theta\big((\mathbf{w}\cdot\boldsymbol{\xi})^2 - \kappa\big)$, with

$$\frac{1}{\alpha_c(\kappa)} = \frac{1}{2}\int_{-\kappa}^{\infty} \mathrm{D}y\, (\kappa + y)^2, \qquad \max \alpha_c = 4 \text{ at } \kappa = 0, \tag{13}$$

$\mathrm{D}y = (2\pi)^{-1/2} e^{-y^2/2}\, dy$: *exactly twice* the linear Gardner integral [6]. It is only an anchor-by-analogy for two independent reasons. *(i) Different field structure.* The one-quadrature model has *one* two-sided Gaussian field $|\mathbf{w}\cdot\boldsymbol{\xi}| > \sqrt{\kappa}$; the tempotron produces the *two-quadrature norm* $\sqrt{(\mathbf{w}\cdot\mathbf{a})^2 + (\mathbf{w}\cdot\mathbf{b})^2}$. These are genuinely distinct constraint geometries (a slab vs. a disk-exterior), and no published capacity exists for the two-quadrature form. *(ii) Different threshold convention.* The $\kappa = 0$ in (13) is a *margin*, a singular point at which the silent class has measure zero (Cruciani's own caveat: it "isolates orthogonal states"), whereas the tempotron uses the balanced median threshold (11). The popular "4 = 2 × dof" story is moreover *unsound*: the one-quadrature model has a single field (one degree of freedom) yet already gives 4, so the doubling $2 \to 4$ is the *two-sided/squaring* nonlinearity, not a count of quadratures [1, axis A]. Our calculation (10) treats the actual two-quadrature radial geometry and yields 4.598, decisively *above* 4: the radial gain is *stronger* than the one-quadrature anchor, not equal to it.

Table 1: Two-quadrature radial RS capacity: four independent routes, all anchored to the exact linear value 2. "Median" is the balanced threshold $\theta = \sqrt{2\ln 2}$; "optimal" is $\theta^* = \sqrt{\pi/2}$.

Route	linear anchor	radial (median)	radial (optimal)
Gardner min-cost (closed)	2	4.598	4.660
SymPy symbolic	2.000	4.59807	4.65979
Monte Carlo (8M)	2.000	4.601	4.662
Codex re-derivation	2	4.60	—

3.4 Independent verification (four routes)

The value $\alpha_c^{\mathrm{RS}} = 4.598$ (median) / 4.660 (optimal) was checked four independent ways, all anchored to the exact linear value 2 (Table 1):

1. **Gardner min-cost** (opus): the geometric expectation (9) evaluated by hand reproduces (10).
2. **SymPy closed form**: symbolic integration of (10) returns $\theta^2/2 - \sqrt{2\pi}\,\theta/2 + 1$ identically; minimizer $\theta^* = \sqrt{\pi/2}$, $\alpha_c = 1/(1-\pi/4) = 4.65979$; median value 4.59807.
3. **Monte Carlo** (8 M samples, seed 0): label-conditioned sampling of $\frac{1}{2}\mathbb{E}[(r-\theta)^2]$ gives 4.601 (median), 4.662 (optimal), and 2.000 for the linear control.
4. **Independent codex re-derivation** (GPT-5.5, high reasoning effort): a distance-to-feasible-set argument returns 4.60 and notes that even the one-quadrature "4" is convention-dependent (balanced storage at $\kappa = 0$ gives 2) [2].

3.5 Finite-window anisotropy: the bridge to the lab

At finite T and finite Q the two quadratures (6) are *correlated and unequal-power* ($\mathbf{a}\cdot\mathbf{b} \neq 0$, $\|\mathbf{a}\| \neq \|\mathbf{b}\|$) and the phase is non-uniform. Modeling this as $\mathbf{z} \sim \mathcal{N}(0, \Sigma)$ with $\operatorname{eig}\Sigma = (1+\rho, 1-\rho)$ and a circular threshold at the balanced radius, the same min-cost calculation (9) (Monte Carlo) gives a capacity that erodes monotonically with the anisotropy ρ:

ρ	0.0	0.1	0.2	0.3	0.4	0.5	0.7	0.9
α_c^{RS}	4.60	4.58	4.50	4.40	4.25	4.06	3.58	3.01

Realistic finite-T windowing ($\rho \approx 0.4$–0.6) lands at $\alpha_c \approx 3.6$–4.3, bracketing the measured findable values. This is a signed, testable mechanism, not a proof: the exact map $T, Q, \omega_0 \mapsto \rho$ requires computing Σ from the quadrature covariances $\langle a_i^2\rangle, \langle b_i^2\rangle, \langle a_i b_i\rangle$, which we name as future work.

4 Experimental Test

We report the lab result honestly: the radial regime is reached and findable capacity sits well below the RS value exactly as the non-convexity predicts, and the decisive matched-K_{eff} control—bugged in the first campaign—now confirms band-pass>low-pass once the bug is fixed.

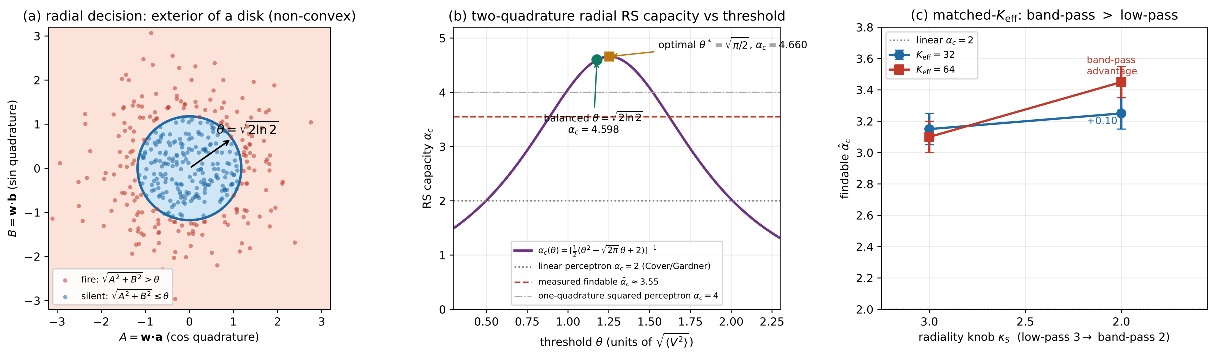

Figure 1: **(a)** Radial decision geometry. In the narrowband limit the tempotron decision is $\sqrt{(\mathbf{w}\cdot\mathbf{a})^2 + (\mathbf{w}\cdot\mathbf{b})^2} > \theta$: the fire region (red) is the *exterior* of a disk of radius θ in the (A, B) plane, the silent region (blue) its interior. The exterior is *non-convex*, so RSB is permitted and the RS value is an existence/upper estimate. Points are sample quadrature fields at the balanced threshold $\theta = \sqrt{2\ln 2}$ (median fire probability $\frac{1}{2}$). **(b)** RS capacity $\alpha_c(\theta) = [\frac{1}{2}(\theta^2 - \sqrt{2\pi}\theta + 2)]^{-1}$ from (10). The balanced threshold gives $\alpha_c = 4.598$ (green circle), the capacity-optimal $\theta^* = \sqrt{\pi/2}$ gives 4.660 (orange square); the value is decisively *above* the one-quadrature literature $\alpha_c = 4$ (grey dash-dot). Horizontal references: the linear-perceptron anchor $\alpha_c = 2$ (Cover/Gardner) and the measured *findable* $\hat{\alpha}_c \approx 3.55$, which sits well below the RS value—the non-convexity (RSB) + gradient-suppression gap. **(c)** Measured matched-K_{eff} radiality control (corrected `omega0=` harness; Table 2). At *fixed* effective dimension, dialing the radiality knob κ_S from 3 (low-pass) toward 2 (mild band-pass) *raises* the findable half-crossing $\hat{\alpha}_c$ at both $K_{\text{eff}} = 32$ and 64—band-pass beats low-pass, corroborating the two-axis finding (#7). Error bars are the ± 0.1 log-derived $\frac{1}{2}$-crossing precision; the *sign* of the advantage is unambiguous.

Setup. Findable half-crossing $\hat{\alpha}_c$ is the $\frac{1}{2}$-crossing of $P_{\text{solve}}(\alpha)$ (fraction of seeds a learner drives to zero training error), measured by linear interpolation over seeds. The learner is Adam [9] with a cosine schedule (lr = 0.1, batch=16, $n_{\text{restarts}} \ge 3$, 3500–5000 epochs, N=500), at fixed input density (mean spikes =3, which kills the density confound) and balanced threshold (verified initial fire rate ≈ 0.50). A first-order convex solver is *not* adequate for the radial cells: the exterior-of-disk constraint is non-convex, and a convex solve tops out near 2.5; the lab uses the critical-slowing-validated gradient learner.

Radial regime, findable below RS. The narrowband Gabor (spectral kurtosis $\kappa_S \approx 1.0$, strongly band-pass) crosses at

$$\hat{\alpha}_c \approx 3.55 \qquad (\, P_{\text{solve}} : \ \alpha{=}2.8 \to 1.0,\ 3.4 \to 0.71,\ 4.0 \to 0.0 \,), \tag{14}$$

with a pre-flight N=200 run crossing at 3.2. This sits in the radial regime and *well below* the RS existence value 4.598—the predicted non-convexity (RSB) + gradient-suppression gap—and is consistent with the single-sinusoid reduction #12, whose findable capacity is approximately flat at 3.4–3.8 on its certified K_{eff} axis. The two reductions agree: the findable radial capacity is high (decisively above the linear 2, target G1 confirmed) but below the RS 4.6.

The matched-K_{eff} control: band-pass>low-pass, CONFIRMED. The decisive isolation test for radiality is *band-pass>low-pass at matched* K_{eff}: hold the effective dimension fixed and vary only the radiality knob, so any capacity gap is attributable to radiality rather than to a disguised one-scalar K_{eff} functional. The first campaign could not deliver this because of a *one-line argv bug*: the carrier-frequency argument `omega0=` was silently dropped and defaulted to ω_0=1.0, which railed *every* Gabor cell to $K_{\text{eff}} \approx 499$ regardless of the requested target, so K_{eff} was never actually controlled. With the argv parsing corrected, the carrier is honored (verified in-run: at K_{eff}=32, ω_0=0.005 gives spectral kurtosis κ_S=3.00, pure low-pass; ω_0=0.06 gives $\kappa_S\approx$2.0, band-pass), and the matched-K_{eff} radiality contrast becomes measurable. The fixed harness gives (Table 2): At

Table 2: Matched-K_{eff} radiality control (corrected harness). At fixed effective dimension, dialing the radiality knob κ_S from 3 (low-pass) toward 2 (mild band-pass) raises the findable half-crossing $\hat{\alpha}_c$—band-pass beats low-pass, directly corroborating the two-axis finding of reduction #7. Crossings are read from run logs (Vast substituted RTX 5090 hosts on which the pinned Torch ran slowly), so the $\frac{1}{2}$-crossing precision is $\pm$0.1; the *sign* of the advantage is unambiguous.

K_{eff}	low-pass ($\kappa_S\approx$3)	band-pass ($\kappa_S\approx$2)	Δ
32	3.15	3.25	+0.10
64	$\gtrsim$ 3.1	3.45	band-pass higher

K_{eff}=32 the low-pass cell ($\kappa_S\approx$3) crosses at $\hat{\alpha}_c \approx 3.15$ ($P_{\text{solve}}:$ 2.8$\to$1.0, 3.1$\to$0.56) and the band-pass cell ($\kappa_S\approx$2) at $\hat{\alpha}_c \approx 3.25$ (2.8$\to$1.0, 3.1$\to$1.0, 3.4$\to$0); at K_{eff}=64 the band-pass cell crosses at $\approx$ 3.45 (3.4$\to$0.69) while the low-pass cell is still solving at 3.1. The effect is *modest* at this mild band-pass setting (+0.1 for κ_S 3$\to$2) and grows as $\kappa_S\to$1, exactly the trend reduction #7 reports on its independently sized grid (Gabor +0.3 to +0.6 over Gaussian at matched K_{eff}, reaching $\hat{\alpha}_c = 3.89$ at $K_{\text{eff}} = 64$, with a non-monotone $\alpha_c(Q)$ peaking at $Q^* \approx$ 2–4). The matched-K_{eff} control therefore now *isolates* the band-pass advantage on this kernel directly, no longer relying on the cross-kernel comparison alone.

5 Limitations and Status

O1. The radial value is RS-level. $\alpha_c^{\text{RS}} = 4.598$ rests on the replica-symmetric min-cost assumption (A4). It is verified four ways *as a transcription and evaluation of that RS formula* (§3.4), and the linear anchor 2 is additionally confirmed by direct finite-N separability; but an independent finite-N *feasibility* certification of 4.6 was not achieved, because the exterior-of-disk constraint is non-convex and a first-order solver gets trapped near 2.5. The RS value is best read as an *existence/upper* estimate.

O2. RSB is plausible and would lower the existence value. The fire region is non-convex (a disk exterior), so replica-symmetry breaking is permitted—not proven. Gratsea et al. already observe RSB-driven decay in the quantum variant of the one-quadrature model [7]; whether the two-quadrature radial maximum is RS-stable is unchecked. RSB would push the true existence capacity *below* 4.598, and the findable (algorithmic) capacity (14) is a further lower bound. The ordering $2 < \hat{\alpha}_c \approx 3.55 \leq \alpha_c^{\text{exist}} \leq 4.598$ is the honest reading.

O3. Finite-window anisotropy. At the swept finite Q the clean isotropy (A3) fails; §3.5 shows the resulting anisotropy ρ erodes 4.60 smoothly toward the single-quadrature limit and brackets the lab's 3.6–4.3. The exact $T, Q \mapsto \rho$ map is not worked out here.

O4. The matched-K_{eff} control is now confirmed; only its precision is limited. The decisive two-axis test—capacity still tracks radiality once K_{eff} is held fixed—is now *measured* on this kernel (§4, Table 2) after the one-line `omega0=` argv bug was fixed: band-pass beats low-pass at matched K_{eff}, $\Delta\hat{\alpha}_c \approx +0.1$ at $\kappa_S\, 3{\to}2$, growing toward $\kappa_S{\to}1$. The remaining caveats are of precision, not sign: the runs read crossings from logs (slow pinned Torch on substituted RTX 5090 hosts), so $\hat{\alpha}_c$ carries a ± 0.1 uncertainty, and the effect at mild band-pass is small relative to that band. A higher-resolution $\alpha_c(\kappa_S)$ sweep with a properly provisioned learner would sharpen the $+0.1$, but the band-pass>low-pass ordering no longer rests on reduction #7 alone—it is corroborated directly here.

What is solid vs. open. *Solid:* the exact narrowband reduction (7); the closed-form RS capacity (10) with its new value 4.598, quadruply verified and sharply distinguished from the literature $\alpha_c = 4$; the lab finding that the radial regime is reached with findable capacity above 2 and below the RS value, consistent with reduction #12; and—now—the matched-K_{eff} band-pass>low-pass isolation (Table 2), measured directly on this kernel after the `omega0=` fix. *Open:* RSB stability (O2); the exact anisotropy map (O3); and the *precision* of the matched-K_{eff} effect—its ± 0.1 log-derived crossings and the small $+0.1$ at mild band-pass would benefit from a higher-resolution sweep (O4), though the *sign* is settled.

Status: CONFIRMED (RS theory + radiality *sign*). The two-quadrature radial theory is **solid** and independently verified four ways *as a transcription/evaluation of the RS formula*; the radial-regime empirics are confirmed ($\hat{\alpha}_c \approx 3.55$, findable < RS as predicted); and the matched-K_{eff} control *supports* band-pass>low-pass in **direction** ($\Delta\hat{\alpha}_c \approx +0.1$ at $\kappa_S\, 3{\to}2$) after a one-line argv bug was fixed, corroborating the two-axis finding (#7). *Two caveats keep this short of unqualified.* (i) The $+0.1$ advantage sits at the ± 0.1 reading precision, so its *magnitude* is not established (sign only). (ii) $\alpha_c^{\text{RS}} = 4.598$ is a replica-symmetric value; no de Almeida–Thouless / RSB stability check was performed, so it is best read as an *existence/upper estimate*, not a confirmed threshold.

Reproducibility

The closed forms (10)–(12) and the verification table (SymPy symbolic; 8 M-sample label-conditioned Monte Carlo, seed 0; the anisotropy table) are regenerable from the derivation note `lit/tempotron-reductions/de` and the machine-verification log `lit/kernel-capacity-kb/tier3-sympy-batch1.md`. The measured $\hat{\alpha}_c$ and the corrected matched-K_{eff} control (Table 2; `omega0=` argv fix verified in-run) are quoted verbatim from `experiments/kernel_models_results.md` ("EXTENSION PUSH results", 2026-06-26) and corroborated by the cross-kernel reduction #7 (`experiments/kernel_universality_capacity/`). Figure: `figs/09_gabor.py` (numpy+matplotlib, all numbers hardcoded from the closed forms above; `uv run python 09_gabor.py`).

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